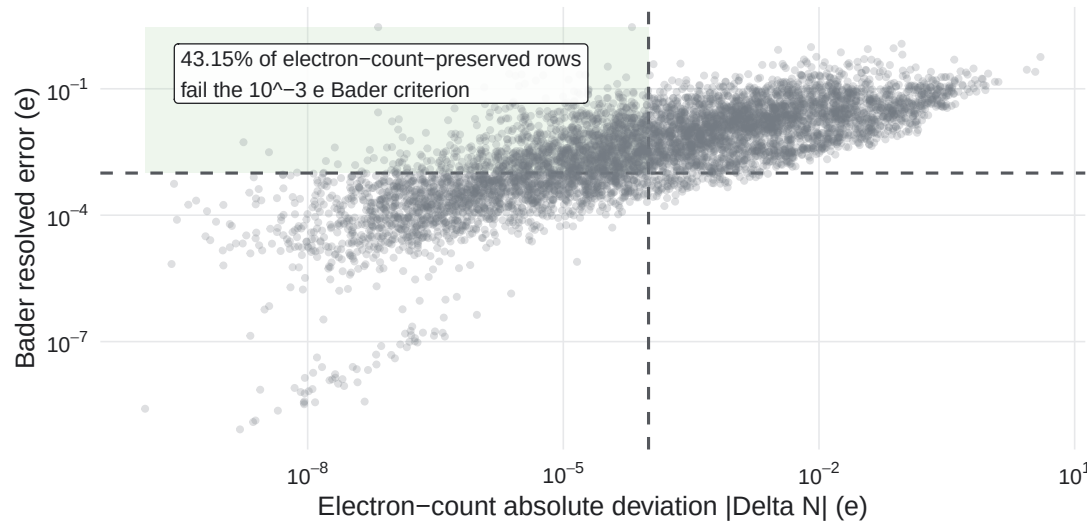


Figure 2 | QoI hierarchy reveals operator-dependent error propagation

global linear: electron count $\rightarrow$ smooth nonlocal: Hartree potential $\rightarrow$ topology-dependent: Bader charge

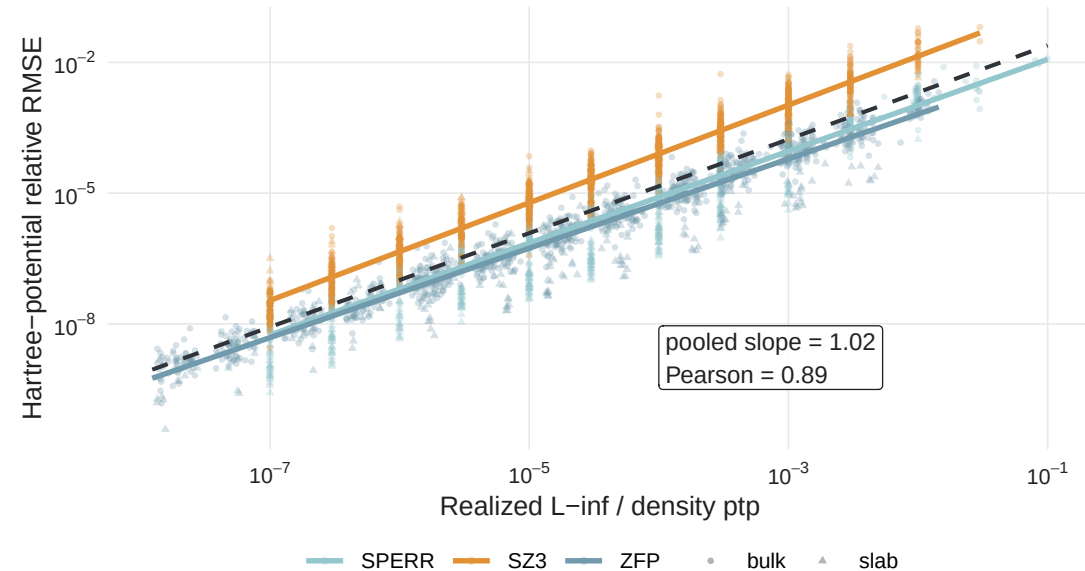
A | Global conservation does not guarantee local fidelity

Same reconstructions; dashed thresholds mark
 $|\Delta N| = 10^{-4}$ e and Bader error = 10^{-3} e



B | Hartree potential follows a smooth near-linear response

All codecs and bulk/slab systems;
dashed line: pooled log-log fit



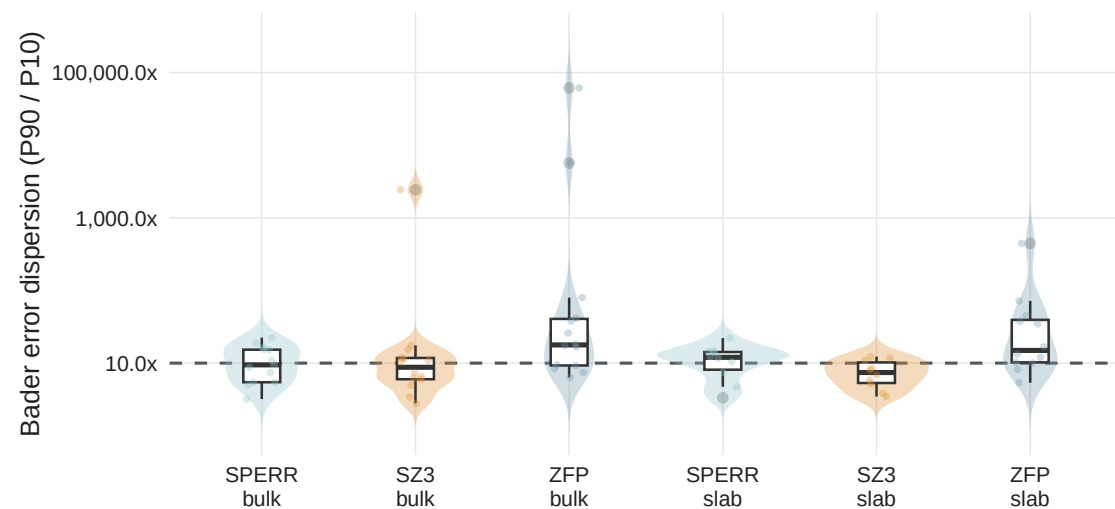
C | Operator structure changes error regularity

Material-level log-log R2; Hartree median 0.994–0.997,
Bader median 0.89–0.95



D | Similar Hartree fidelity does not imply similar Bader fidelity

P90/P10 spread within 0.5-decade Hartree-error bins;
55.4% of rows lie in bins with at least 10x spread



Hartree statistics use reproduction-gate-passing rows; 73 SZ3 stream-size mismatches remain excluded by the frozen gate.

Slab conclusions rely on R2, local elasticity and jump magnitude rather than universal rung-wise monotonicity.